

Monomial Exponential Series

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Contents

1	Introduction	3
2	Isolating the Basic Blocks	3
3	Connection with Ordinary Monomials	4
4	Composition by Powers	5
5	Composition Operators	5
6	Prime Factorisation as Operator Factorisation	6
7	Divisibility as Reachability	7
8	A Divisibility Diagram	8
9	Greatest Common Divisors and Least Common Multiples	8
10	Prime Blocks are Indecomposable	9
11	Composition Acting on the Coefficients	10
12	Dirichlet Convolution Appears	11
13	The Divisor-Sum Operator	12
14	Möbius Inversion	13
15	Changing the Exponential Base	13
16	Taylor Expansion of a Block	14
17	Taylor Coefficients and Divisors	15
18	Centered Blocks	15
19	A New Expansion in Centered Blocks	16
20	Formal Representation Theorem	17

21 Prime Indices in the Taylor Transform	18
22 Linear Independence	18
23 Differentiation	19
24 Connection with Differential Equations	20
25 Multiplication of Blocks	20
26 Convergence	21
26.1 The Region $ x < 1$	21
26.2 The Point $x = 1$	22
26.3 The Point $x = 0$	22
26.4 The Region $x > 1$ with $b > 1$	23
26.5 The Region $x > 1$ with $0 < b < 1$	23
26.6 Negative Values with $x < -1$	24
27 A Simple Example	25
28 Complex Extension	25
29 What is Inherited and What May Be Interesting	26
30 Research Directions	26
31 Where We Are	27

1 Introduction

We begin with the family of series

$$F(x) = a_0 + a_1b^{x^2} + a_2b^{x^3} + a_3b^{x^4} + \cdots.$$

A slightly cleaner indexing is obtained by writing

$$F(x) = a_0 + \sum_{n=1}^{\infty} c_n b^{x^n}.$$

Throughout this text we assume

$$b > 0, \quad b \neq 1.$$

It will often be useful to introduce

$$\lambda = \ln b.$$

Since

$$b^{x^n} = e^{(\ln b)x^n},$$

the series can equivalently be written as

$$F(x) = a_0 + \sum_{n=1}^{\infty} c_n e^{\lambda x^n}.$$

We shall provisionally call such an expression a *monomial exponential series*.

The purpose of this document is not to claim that this family of functions has never appeared before, but rather to investigate the structures that naturally arise from it.

In particular, we shall see connections with

composition $\longleftrightarrow$ multiplication of indices,

divisibility $\longleftrightarrow$ reachability between blocks,

and, surprisingly,

composition operators $\longleftrightarrow$ Dirichlet convolution.

2 Isolating the Basic Blocks

Let us forget the coefficients for a moment and isolate the functions

$$\phi_n(x) = b^{x^n}.$$

Equivalently,

$$\phi_n(x) = e^{\lambda x^n}.$$

The first blocks are

$$\phi_1(x) = b^x,$$

$$\phi_2(x) = b^{x^2},$$

$$\phi_3(x) = b^{x^3},$$

$$\phi_4(x) = b^{x^4},$$

and so on.

Our series is therefore a linear combination of these blocks:

$$F(x) = a_0 + \sum_{n=1}^{\infty} c_n \phi_n(x).$$

Thus the family

$$\mathcal{B} = \{1, \phi_1, \phi_2, \phi_3, \dots\}$$

plays a role analogous to the familiar monomial family

$$\{1, x, x^2, x^3, \dots\}.$$

However, its algebraic and analytic behaviour is very different.

3 Connection with Ordinary Monomials

The blocks are obtained directly from the ordinary monomials.

Let

$$q_n(x) = x^n.$$

Then

$$\phi_n(x) = b^{q_n(x)}.$$

Therefore

$$\phi_n = \exp_b \circ q_n,$$

where

$$\exp_b(y) = b^y.$$

This observation is important because some of the structure we shall discover is inherited from the monomials themselves.

For example,

$$q_n(q_m(x)) = (x^m)^n = x^{mn}.$$

Thus

$$\boxed{q_n \circ q_m = q_{mn}.$$

The monomials already encode multiplication of their indices through composition. The exponential blocks preserve this structure after exponentiation. This distinction will be important:

The multiplicative behaviour of the indices is inherited from the monomials x^n , while the new questions arise from taking linear combinations of the exponentiated monomials b^{x^n} .

4 Composition by Powers

Consider

$$\phi_n(x) = b^{x^n}.$$

Replace x by x^m . Then

$$\phi_n(x^m) = b^{(x^m)^n} = b^{x^{mn}}.$$

Therefore

$$\boxed{\phi_n(x^m) = \phi_{mn}(x).$$

For example,

$$\phi_2(x^3) = b^{(x^3)^2} = b^{x^6} = \phi_6(x),$$

while

$$\phi_3(x^2) = b^{(x^2)^3} = b^{x^6} = \phi_6(x).$$

Hence

$$\boxed{\phi_2(x^3) = \phi_3(x^2) = \phi_6(x).$$

More generally,

$$\boxed{\phi_n(x^m) = \phi_m(x^n) = \phi_{mn}(x).$$

Thus multiplication of natural-number indices corresponds to composition with power maps.

5 Composition Operators

The previous phenomenon becomes clearer if we introduce an operator.

For every $m \in \mathbb{N}$, define

$$\boxed{T_m F(x) = F(x^m).$$

Then, when T_m acts on one of our basic blocks,

$$T_m \phi_n(x) = \phi_n(x^m).$$

Therefore

$$\boxed{T_m \phi_n = \phi_{mn}.$$

Now consider two such operators:

$$T_m T_n F(x).$$

We first apply T_n :

$$T_n F(x) = F(x^n).$$

Then

$$T_m T_n F(x) = F((x^m)^n) = F(x^{mn}).$$

Hence

$$\boxed{T_m T_n = T_{mn}.$$

Since multiplication of natural numbers is commutative,

$$mn = nm,$$

we also have

$$\boxed{T_m T_n = T_n T_m.$$

Finally,

$$T_1 F(x) = F(x),$$

so

$$\boxed{T_1 = I.$$

We therefore obtain the same algebraic laws as the multiplicative monoid

$$(\mathbb{N}, \cdot).$$

6 Prime Factorisation as Operator Factorisation

Suppose

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$$

is the prime factorisation of n .

Because

$$T_m T_n = T_{mn},$$

we obtain

$$T_n = T_{p_1}^{\alpha_1} T_{p_2}^{\alpha_2} \dots T_{p_r}^{\alpha_r}.$$

For example,

$$12 = 2^2 \cdot 3,$$

and therefore

$$T_{12} = T_2^2 T_3.$$

Acting on ϕ_1 ,

$$T_{12}\phi_1 = \phi_{12}.$$

But we may reach the same block through

$$T_2 T_2 T_3 \phi_1.$$

Thus prime factorisation of integers becomes factorisation of composition operators. This suggests that the operators

$$T_2, T_3, T_5, T_7, \dots$$

play the role of fundamental generators.

7 Divisibility as Reachability

We now obtain a direct interpretation of divisibility.

Suppose $d, n \in \mathbb{N}$.

If

$$d \mid n,$$

then there exists an integer m such that

$$n = dm.$$

Consequently,

$$T_m \phi_d = \phi_{dm} = \phi_n.$$

Therefore

$$d \mid n \iff \phi_n \text{ can be reached from } \phi_d \text{ by some } T_m.$$

This allows us to interpret the divisibility relation geometrically as a network of function transformations.

For example,

$$\phi_2 \longrightarrow \phi_4 \longrightarrow \phi_8$$

corresponds to

$$2 \mid 4 \mid 8.$$

Similarly,

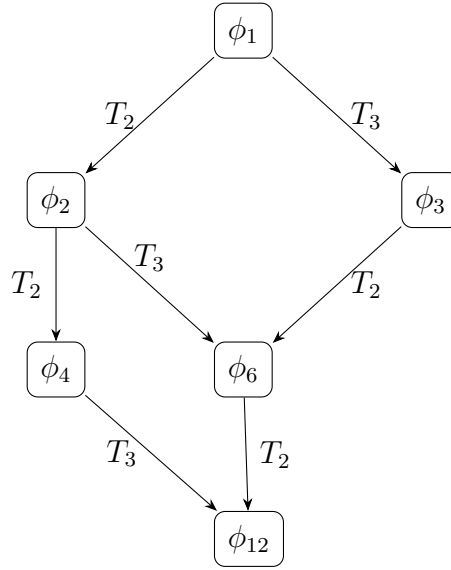
$$\phi_3 \longrightarrow \phi_6 \longrightarrow \phi_{12}$$

corresponds to

$$3 \mid 6 \mid 12.$$

8 A Divisibility Diagram

The relation can be visualised using a small portion of the divisibility lattice.



Different paths may lead to the same function because multiplication is commutative. For example,

$$T_2 T_3 \phi_1 = T_3 T_2 \phi_1 = \phi_6.$$

9 Greatest Common Divisors and Least Common Multiples

The divisibility interpretation also introduces naturally the greatest common divisor and least common multiple.

Take two blocks

$$\phi_m \quad \text{and} \quad \phi_n.$$

A block ϕ_d is a common predecessor when

$$d \mid m \quad \text{and} \quad d \mid n.$$

The largest possible such index is

$$d = \gcd(m, n).$$

Thus

$$\boxed{\phi_{\gcd(m, n)}}$$

is the greatest common predecessor of ϕ_m and ϕ_n .
Similarly, a block ϕ_k is a common descendant if

$$m \mid k \quad \text{and} \quad n \mid k.$$

The smallest such index is

$$k = \text{lcm}(m, n).$$

Hence

$$\boxed{\phi_{\text{lcm}(m, n)}}$$

is the smallest common descendant.
For example,

$$\gcd(4, 6) = 2$$

and

$$\text{lcm}(4, 6) = 12.$$

Therefore

$$\phi_2$$

is the greatest common predecessor of

$$\phi_4, \phi_6,$$

while

$$\phi_{12}$$

is their smallest common descendant.

10 Prime Blocks are Indecomposable

Consider

$$\phi_6(x) = b^{x^6}.$$

Because

$$6 = 2 \cdot 3,$$

we can write

$$\phi_6(x) = \phi_2(x^3) = \phi_3(x^2).$$

Similarly,

$$12 = 2 \cdot 6 = 3 \cdot 4,$$

so

$$\phi_{12}(x) = \phi_2(x^6) = \phi_3(x^4) = \phi_4(x^3) = \phi_6(x^2).$$

However, suppose p is prime.

If

$$\phi_p(x) = \phi_m(x^n)$$

with

$$m > 1, \quad n > 1,$$

then we would have

$$p = mn,$$

contradicting the primality of p .

Therefore prime-indexed blocks are indecomposable under non-trivial power composition.

Proposition 10.1. *If p is prime, then there do not exist $m, n > 1$ such that*

$$\phi_p(x) = \phi_m(x^n).$$

11 Composition Acting on the Coefficients

Now consider

$$F(x) = \sum_{n \geq 1} c_n \phi_n(x).$$

Applying T_m , we obtain

$$T_m F(x) = F(x^m).$$

Therefore

$$F(x^m) = \sum_{n \geq 1} c_n \phi_n(x^m).$$

Using

$$\phi_n(x^m) = \phi_{mn}(x),$$

we obtain

$$\boxed{T_m F = \sum_{n \geq 1} c_n \phi_{mn} .}$$

For $m = 2$,

$$T_2 F = c_1 \phi_2 + c_2 \phi_4 + c_3 \phi_6 + c_4 \phi_8 + \cdots .$$

Thus the coefficient positions

$$1, 2, 3, 4, \dots$$

are sent to

$$2, 4, 6, 8, \dots$$

For $m = 3$, they are sent to

$$3, 6, 9, 12, \dots$$

More generally,

$$\boxed{n \longmapsto mn .}$$

12 Dirichlet Convolution Appears

A stronger connection appears when we combine several composition operators.

Let

$$A = \sum_{m \geq 1} d_m T_m ,$$

where d_m is another sequence.

Apply A to

$$F(x) = \sum_{n \geq 1} c_n \phi_n(x) .$$

Then

$$AF = \sum_{m \geq 1} d_m T_m \left(\sum_{n \geq 1} c_n \phi_n \right) .$$

Formally,

$$AF = \sum_{m, n \geq 1} d_m c_n \phi_{mn} .$$

Now collect all terms whose index is equal to k .

The equation

$$mn = k$$

is equivalent to

$$m \mid k, \quad n = \frac{k}{m}.$$

Therefore the coefficient of ϕ_k becomes

$$\sum_{m \mid k} d_m c_{k/m}.$$

Thus

$$AF = \sum_{k \geq 1} \left(\sum_{m \mid k} d_m c_{k/m} \right) \phi_k.$$

But the expression

$$(d * c)(k) = \sum_{m \mid k} d_m c_{k/m}$$

is precisely the *Dirichlet convolution* of the sequences d and c .
Therefore

$$\left(\sum_{m \geq 1} d_m T_m \right) F \longleftrightarrow d * c.$$

This gives a direct connection between composition of functions and one of the central operations of multiplicative number theory.

13 The Divisor-Sum Operator

Consider the special case

$$d_m = 1$$

for every m .

Define formally

$$D = \sum_{m \geq 1} T_m.$$

Then the coefficient of ϕ_k in DF is

$$\sum_{m \mid k} c_{k/m}.$$

Renaming the divisor,

$$\sum_{d \mid k} c_d.$$

Thus the operator D transforms the coefficient sequence

$$c_k$$

into its divisor-sum sequence

$$g_k = \sum_{d|k} c_d.$$

This is exactly the type of transformation for which Möbius inversion is designed.

14 Möbius Inversion

Suppose

$$g_n = \sum_{d|n} c_d.$$

Then Möbius inversion gives

$$c_n = \sum_{d|n} \mu(d) g_{n/d},$$

where μ is the Möbius function.

Therefore a classical number-theoretic inversion formula appears naturally from the composition operators of our function family.

Symbolically,

$$\mathbf{1} * c = g,$$

and

$$\mu * g = c,$$

because

$$\mu * \mathbf{1} = \varepsilon.$$

Here ε is the identity for Dirichlet convolution.

This suggests a possible bridge between monomial exponential series and arithmetic functions.

15 Changing the Exponential Base

The choice of base b can be simplified.

Since

$$b = e^\lambda,$$

we have

$$b^{x^n} = e^{\lambda x^n}.$$

Therefore it is enough to study

$$\phi_n(x) = e^{\lambda x^n}.$$

The parameter

$$\lambda = \ln b$$

contains all information about the base.

There are three fundamentally different cases:

$$\lambda > 0 \iff b > 1,$$

$$\lambda < 0 \iff 0 < b < 1,$$

and

$$\lambda = 0 \iff b = 1.$$

The case $b = 1$ is trivial because

$$1^{x^n} = 1$$

for every n .

We shall therefore exclude it.

16 Taylor Expansion of a Block

The representation

$$\phi_n(x) = e^{\lambda x^n}$$

reveals another important structure.

Recall that

$$e^y = \sum_{k=0}^{\infty} \frac{y^k}{k!}.$$

Substituting

$$y = \lambda x^n,$$

we obtain

$$e^{\lambda x^n} = \sum_{k=0}^{\infty} \frac{\lambda^k x^{nk}}{k!}.$$

Therefore

$$\phi_n(x) = 1 + \lambda x^n + \frac{\lambda^2}{2!} x^{2n} + \frac{\lambda^3}{3!} x^{3n} + \dots$$

Notice something important:

$$\phi_n$$

contains only powers whose exponents are multiples of n :

$$n, 2n, 3n, 4n, \dots$$

Thus divisibility appears once again.

17 Taylor Coefficients and Divisors

Consider formally

$$F(x) = \sum_{n \geq 1} c_n e^{\lambda x^n}.$$

Using the Taylor expansion,

$$F(x) = \sum_{n \geq 1} c_n \sum_{k \geq 0} \frac{\lambda^k}{k!} x^{nk}.$$

Suppose we want the coefficient of x^m .

A contribution appears exactly when

$$nk = m.$$

Therefore

$$n \mid m.$$

For $m \geq 1$, the coefficient of x^m is consequently

$$[x^m]F(x) = \sum_{n \mid m} c_n \frac{\lambda^{m/n}}{(m/n)!}.$$

This is another divisor sum.

Thus divisibility enters the theory in two completely different ways:

composition and Taylor expansion.

This repetition suggests that the divisor structure is not merely accidental.

18 Centered Blocks

There is one inconvenience with the blocks

$$\phi_n(x) = e^{\lambda x^n}.$$

At $x = 0$,

$$\phi_n(0) = 1$$

for every n .

It is therefore useful to define the centered blocks

$$\boxed{\psi_n(x) = e^{\lambda x^n} - 1.}$$

Now

$$\psi_n(0) = 0.$$

Their Taylor expansion is

$$\psi_n(x) = \lambda x^n + \frac{\lambda^2}{2!} x^{2n} + \frac{\lambda^3}{3!} x^{3n} + \cdots .$$

The lowest-degree term is exactly

$$\boxed{\lambda x^n.}$$

This gives the family a triangular relationship with the ordinary monomials.

19 A New Expansion in Centered Blocks

Consider an ordinary formal power series

$$f(x) = f(0) + t_1 x + t_2 x^2 + t_3 x^3 + \cdots .$$

We can ask whether it is possible to write

$$\boxed{f(x) = f(0) + \sum_{n \geq 1} c_n (e^{\lambda x^n} - 1) .}$$

Expanding the right-hand side gives

$$\sum_{n \geq 1} c_n \sum_{k \geq 1} \frac{\lambda^k}{k!} x^{nk} .$$

Therefore the coefficient t_m must satisfy

$$\boxed{t_m = \sum_{n|m} c_n \frac{\lambda^{m/n}}{(m/n)!} .}$$

Among the divisors of m is $n = m$.

The corresponding contribution is

$$c_m \lambda.$$

Hence

$$t_m = \lambda c_m + \sum_{\substack{n|m \\ n < m}} c_n \frac{\lambda^{m/n}}{(m/n)!} .$$

Solving for c_m ,

$$c_m = \frac{1}{\lambda} \left[t_m - \sum_{\substack{n|m \\ n < m}} c_n \frac{\lambda^{m/n}}{(m/n)!} \right].$$

This gives a recursive procedure for constructing the coefficients c_m .

20 Formal Representation Theorem

The previous recursion gives an interesting result.

Theorem 20.1. *Let*

$$\lambda \neq 0.$$

Every formal power series

$$f(x) = f(0) + \sum_{m \geq 1} t_m x^m$$

has a unique formal representation of the form

$$f(x) = f(0) + \sum_{n \geq 1} c_n (e^{\lambda x^n} - 1).$$

Proof. For $m = 1$,

$$t_1 = \lambda c_1,$$

so

$$c_1 = \frac{t_1}{\lambda}.$$

Suppose

$$c_1, \dots, c_{m-1}$$

have already been determined.

Then

$$t_m = \lambda c_m + \sum_{\substack{n|m \\ n < m}} c_n \frac{\lambda^{m/n}}{(m/n)!}.$$

Every term in the sum is already known.

Since

$$\lambda \neq 0,$$

we can solve uniquely for c_m .

Proceeding recursively determines every coefficient c_m .

□

Remark 20.1. *This theorem concerns formal power series.*

It does not automatically imply that the corresponding infinite series converges as a function for a particular value of x .

Understanding when the formal representation becomes an analytic representation is a separate and important problem.

21 Prime Indices in the Taylor Transform

The divisor formula becomes especially simple for prime indices.

Suppose p is prime.

Its only positive divisors are

$$1 \quad \text{and} \quad p.$$

Therefore

$$t_p = c_1 \frac{\lambda^p}{p!} + c_p \lambda.$$

Hence

$$c_p = \frac{1}{\lambda} \left(t_p - c_1 \frac{\lambda^p}{p!} \right).$$

For a composite number, more terms appear.

For example, the divisors of 6 are

$$1, 2, 3, 6.$$

Therefore

$$t_6 = c_1 \frac{\lambda^6}{6!} + c_2 \frac{\lambda^3}{3!} + c_3 \frac{\lambda^2}{2!} + c_6 \lambda.$$

Thus the arithmetic complexity of the index controls the number of interactions between coefficients.

22 Linear Independence

The centered blocks are also linearly independent.

Proposition 22.1. *For $\lambda \neq 0$, the functions*

$$\psi_n(x) = e^{\lambda x^n} - 1, \quad n \geq 1,$$

are linearly independent with respect to finite linear combinations.

Proof. Suppose

$$c_1 \psi_1(x) + c_2 \psi_2(x) + \cdots + c_N \psi_N(x) = 0.$$

Assume not all coefficients are zero.

Let m be the smallest index for which

$$c_m \neq 0.$$

The lowest power of x contributed by ψ_m is

$$\lambda x^m.$$

Blocks with index greater than m cannot contribute to x^m .

Blocks with index smaller than m have coefficient zero by the minimality of m .

Therefore the coefficient of x^m in the entire expression is

$$\lambda c_m.$$

Since

$$\lambda \neq 0 \quad \text{and} \quad c_m \neq 0,$$

this coefficient cannot vanish.

This is a contradiction.

Hence every c_n must be zero.

□

23 Differentiation

Let

$$\phi_n(x) = e^{\lambda x^n}.$$

Differentiating gives

$$\phi'_n(x) = n\lambda x^{n-1} e^{\lambda x^n}.$$

Therefore

$$\boxed{\phi'_n(x) = n\lambda x^{n-1} \phi_n(x).}$$

The original family

$$\{\phi_n\}$$

is therefore not closed under differentiation because a polynomial factor appears.

However, this suggests enlarging the family.

Consider functions of the form

$$x^k \phi_n(x).$$

Differentiate:

$$\frac{d}{dx} (x^k \phi_n(x)) = kx^{k-1} \phi_n(x) + x^k \phi'_n(x).$$

Therefore

$$\frac{d}{dx}(x^k \phi_n(x)) = kx^{k-1} \phi_n(x) + n\lambda x^{k+n-1} \phi_n(x).$$

Both terms remain in the enlarged family

$$\mathcal{D} = \{x^k e^{\lambda x^n} : k \geq 0, n \geq 1\}.$$

Thus this larger family *is* closed under differentiation.

24 Connection with Differential Equations

Each block satisfies a first-order differential equation.

From

$$\phi'_n(x) = n\lambda x^{n-1} \phi_n(x),$$

we obtain

$$\phi'_n(x) - n\lambda x^{n-1} \phi_n(x) = 0.$$

Define the differential operator

$$L_n = \frac{d}{dx} - n\lambda x^{n-1}.$$

Then

$$L_n \phi_n = 0.$$

Thus every basic block is the solution of a simple first-order linear differential equation.

This gives another possible direction:

Research Question 24.1. *Can properties of a monomial exponential series be described using the collection of differential operators*

$$L_1, L_2, L_3, \dots?$$

25 Multiplication of Blocks

Let us also investigate multiplication.

Take

$$\phi_m(x) = e^{\lambda x^m}$$

and

$$\phi_n(x) = e^{\lambda x^n}.$$

Then

$$\phi_m(x) \phi_n(x) = e^{\lambda x^m} e^{\lambda x^n}.$$

Therefore

$$\phi_m(x)\phi_n(x) = e^{\lambda(x^m+x^n)}.$$

In general this is not another basic block.

Thus

$$\{\phi_n\}$$

is not closed under multiplication.

However, multiplication naturally leads to the larger family

$$e^{\lambda P(x)},$$

where $P(x)$ is a polynomial.

For example,

$$\phi_2(x)\phi_5(x) = e^{\lambda(x^2+x^5)}.$$

This suggests that the monomial exponential blocks sit naturally inside a larger theory of exponentials of polynomials.

26 Convergence

We now investigate

$$F(x) = \sum_{n=1}^{\infty} c_n b^{x^n}.$$

The behaviour depends strongly on both x and b .

26.1 The Region $|x| < 1$

If

$$|x| < 1,$$

then

$$x^n \longrightarrow 0.$$

Consequently,

$$b^{x^n} \longrightarrow 1.$$

Thus

$$c_n b^{x^n} \sim c_n.$$

In particular, if

$$c_n = 1,$$

then

$$b^{x^n} \longrightarrow 1,$$

so the terms do not approach zero and

$$\sum_{n=1}^{\infty} b^{x^n}$$

diverges.

More generally, in the region $|x| < 1$, convergence is controlled primarily by the coefficient sequence c_n .

26.2 The Point $x = 1$

At

$$x = 1,$$

we have

$$x^n = 1.$$

Therefore

$$F(1) = \sum_{n=1}^{\infty} c_n b.$$

Hence

$$F(1) = b \sum_{n=1}^{\infty} c_n.$$

Thus convergence at $x = 1$ is exactly the same as convergence of

$$\sum c_n.$$

26.3 The Point $x = 0$

At

$$x = 0,$$

we have

$$0^n = 0.$$

Thus

$$b^{0^n} = 1$$

and

$$F(0) = \sum_{n=1}^{\infty} c_n.$$

Again, convergence is determined directly by the coefficients.

26.4 The Region $x > 1$ with $b > 1$

Suppose

$$x > 1$$

and

$$b > 1.$$

Then

$$x^n \longrightarrow +\infty.$$

Therefore

$$b^{x^n} \longrightarrow +\infty.$$

If $c_n = 1$, the terms themselves grow without bound.
For convergence we require at least

$$c_n b^{x^n} \longrightarrow 0.$$

Thus the coefficients must decay extremely rapidly.

26.5 The Region $x > 1$ with $0 < b < 1$

Now suppose

$$0 < b < 1.$$

Then

$$b^{x^n} \longrightarrow 0$$

extremely rapidly.
Consider the ratio

$$\frac{b^{x^{n+1}}}{b^{x^n}} = b^{x^{n+1} - x^n}.$$

Since

$$x^{n+1} - x^n = x^n(x - 1),$$

we obtain

$$\frac{b^{x^{n+1}}}{b^{x^n}} = b^{x^n(x-1)}.$$

Because

$$x^n(x-1) \rightarrow \infty$$

and $0 < b < 1$,

$$b^{x^n(x-1)} \rightarrow 0.$$

Therefore

$$\sum_{n=1}^{\infty} b^{x^n}$$

converges for every $x > 1$.

The decay is faster than ordinary geometric decay.

26.6 Negative Values with $x < -1$

Negative values introduce a new phenomenon.

Let

$$x = -r, \quad r > 1.$$

Then

$$x^n = (-1)^n r^n.$$

Therefore, for even n ,

$$x^n = r^n,$$

while for odd n ,

$$x^n = -r^n.$$

Hence

$$b^{x^n} = \begin{cases} b^{r^n}, & n \text{ even,} \\ b^{-r^n}, & n \text{ odd.} \end{cases}$$

If

$$b > 1,$$

the even-indexed blocks explode while the odd-indexed blocks decay.

If

$$0 < b < 1,$$

the opposite happens.

Thus one parity tends to zero while the other tends to infinity.

This creates a strong asymmetry that does not occur for $x > 0$.

27 A Simple Example

Consider

$$b = \frac{1}{2}$$

and

$$x = 2.$$

Then

$$b^{x^n} = \left(\frac{1}{2}\right)^{2^n}.$$

The first terms are

$$\frac{1}{2^2}, \quad \frac{1}{2^4}, \quad \frac{1}{2^8}, \quad \frac{1}{2^{16}}, \quad \frac{1}{2^{32}}, \dots$$

or

$$\frac{1}{4}, \quad \frac{1}{16}, \quad \frac{1}{256}, \quad \frac{1}{65536}, \dots$$

The exponent itself grows exponentially:

$$2^n,$$

and this exponential quantity appears inside another exponential.
This explains the extremely rapid decay.

28 Complex Extension

Another possible generalisation is to allow

$$\lambda \in \mathbb{C}.$$

If

$$\lambda = i\theta,$$

then

$$\phi_n(x) = e^{i\theta x^n}.$$

Since

$$|e^{i\theta x^n}| = 1,$$

the blocks no longer grow or decay.
Instead they oscillate.
Thus the same structure becomes

$$F(x) = \sum_{n \geq 1} c_n e^{i\theta x^n}.$$

Now the problem is no longer dominated by exponential growth but by cancellation and oscillation.

This creates a possible connection with polynomial-phase exponential functions and harmonic analysis.

29 What is Inherited and What May Be Interesting

The identity

$$\phi_n(x^m) = \phi_{nm}(x)$$

is ultimately inherited from

$$(x^m)^n = x^{mn}.$$

Therefore this identity by itself should not be considered a new phenomenon.

However, several richer structures appear after the blocks are placed inside linear combinations.

For example,

$$F(x) = \sum c_n \phi_n(x)$$

allows the composition operators T_m to transform entire coefficient sequences.

Combining these operators produces Dirichlet convolution.

Taylor expansion produces divisor sums independently.

The centered family

$$e^{\lambda x^n} - 1$$

also gives a triangular transformation of ordinary power series.

These interactions appear to be more mathematically substantial than the original identity alone.

30 Research Directions

Several natural questions arise from the preceding calculations.

Research Question 30.1. *For which coefficient sequences c_n does*

$$\sum_{n=1}^{\infty} c_n e^{\lambda x^n}$$

converge on a prescribed interval?

Research Question 30.2. *What is the maximal domain of convergence associated with a given sequence c_n ?*

Research Question 30.3. *When does the formal representation*

$$f(x) = f(0) + \sum_{n \geq 1} c_n (e^{\lambda x^n} - 1)$$

converge to the analytic function $f(x)$?

Research Question 30.4. *What properties of f correspond to simple arithmetic properties of the coefficient sequence c_n ?*

Research Question 30.5. *What happens when c_n is itself an arithmetic function such as*

$$\mu(n), \quad \varphi(n), \quad d(n), \quad \Lambda(n)?$$

Research Question 30.6. *Can Dirichlet-series identities be translated into identities between operators of the form*

$$\sum_{m \geq 1} d_m T_m?$$

Research Question 30.7. *Can multiplication, differentiation and composition be combined into a natural algebra generated by the functions*

$$e^{\lambda x^n}?$$

Research Question 30.8. *Are there interesting functional equations involving prime-indexed blocks*

$$\phi_p(x) = e^{\lambda x^p}?$$

Research Question 30.9. *Can analytic properties of*

$$F(x) = \sum c_n e^{\lambda x^n}$$

reveal arithmetic properties of the sequence c_n ?

31 Where We Are

We began with the apparently simple family

$$\phi_n(x) = e^{\lambda x^n}.$$

Composition revealed

$$T_m \phi_n = \phi_{mn}.$$

This transformed multiplication of natural numbers into composition of functions. Divisibility became reachability:

$$d \mid n \iff \phi_d \longrightarrow \phi_n.$$

Prime factorisation became factorisation of composition operators:

$$T_n = \prod_{p^\alpha \parallel n} T_p^\alpha.$$

Combining composition operators led naturally to Dirichlet convolution:

$$(d * c)(n) = \sum_{m|n} d_m c_{n/m}.$$

Taylor expansion independently produced divisor sums:

$$[x^m]F(x) = \sum_{n|m} c_n \frac{\lambda^{m/n}}{(m/n)!}.$$

Finally, after centering the blocks,

$$\psi_n(x) = e^{\lambda x^n} - 1,$$

we discovered that they form a triangular deformation of the ordinary monomials and formally allow every power series to be expanded uniquely in this new family.

Thus the main object is no longer merely the expression

$$\sum c_n e^{\lambda x^n}.$$

The emerging structure is the interaction between

analysis + composition + divisibility + arithmetic functions.

The next step should be to determine which of these formal structures survive analytically when convergence is imposed.